

Advanced Packaging & Substrate Bottleneck Factor

Capturing Value Migration from Front-End Compute
into Constrained Packaging, Substrate, Test & Materials Enablers
Mathematical Framework & Implementation Specification

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Abstract

This document specifies a thematic equity factor designed to capture reratings as AI chips shift value out of front-end compute and into the constrained advanced-packaging supply chain. CoWoS/CoPoS capacity, OSAT outsourcing, wafer/package test, ABF and SAP substrates, CPO packaging, glass/TGV yield, embedded passives, and specialty packaging materials are the binding constraints as larger AI GPUs, ASICs, HBM stacks, and optical packages drive a multi-year packaging bottleneck. The best expression is not “long semis” or “long AI”—it is a targeted trade on the packaging, substrate, test, and materials enablers that turn AI silicon into deployable systems. Scoring combines SEC filing text signals, patent intensity, capacity/backlog language, and margin trajectory into a composite exposure score. Universe: US-listed equities in Semiconductors & Semiconductor Equipment, Electronic Equipment/Components, and Specialty Chemicals. Overall confidence: 0.78–0.82.

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1 Executive Summary

1.1 Factor Theme

Capture reratings as AI chips shift value out of front-end compute and into the constrained advanced-packaging supply chain. The factor targets:

- CoWoS/CoPoS capacity and OSAT outsourcing
- Wafer-level and package test services
- ABF and SAP substrates for AI GPU/ASIC/HBM packaging
- CPO (co-packaged optics) module integration
- Glass core/TGV yield and embedded passives
- Specialty packaging materials (underfill, mold compound, flux, bonding films)

1.2 Investment Rationale

As AI GPUs/ASICs and HBM stacks scale, value migrates from front-end compute into constrained packaging, substrate, test, and materials nodes. Firms with:

- Higher advanced-packaging revenue mix
- Explicit capacity tightness (sold-out utilization, allocation language)
- Substrate complexity exposure (high layer-count ABF/SAP, glass/TGV)
- Test-intensity leverage (growing test time per AI device)
- Next-gen packaging optionality (glass/TGV, CPO, hybrid bonding R&D)
- Margin expansion from mix shift away from legacy services

should rerate as scarcity confers pricing power and margin uplift.

1.3 Target Industry Sectors

Sector / Industry	Coverage
IT — Semiconductors & Equipment	OSAT providers, test equipment, packaging equipment, CoWoS tooling, inspection/metrology
IT — Electronic Equipment & Components	IC substrate manufacturers (ABF, SAP, glass core), embedded passives, CPO module integrators
Materials — Specialty Chemicals	Underfill, mold compound, flux, bonding film, dielectric layers

Table 1: Target industry mappings

1.4 Role Buckets

Each eligible company is tagged with one of:

- **osat**: Advanced packaging, assembly, test (OSAT)
- **substrates**: ABF, SAP, glass core substrate manufacturers
- **test_equipment**: Wafer-level and final test equipment
- **packaging_equipment**: Bonding, inspection, CoWoS tooling
- **embedded_passives**: Embedded passive component makers
- **cpo_module**: Co-packaged optics module integrators
- **specialty_materials**: Underfill, mold compound, flux, bonding films

1.5 Avoid List

- Pure-play front-end foundries without packaging/OSAT operations
- Companies where AI packaging revenue is immaterial or undifferentiated (<10% of sales)
- Names trading at peak-cycle multiples with no incremental capacity coming online
- Commodity materials providers without product differentiation
- SPACs <12 months post-deSPAC; ADRs with restricted liquidity

2 Signal Taxonomy (A–G)

2.1 Signal A: Advanced-Packaging Revenue Concentration

Definition: Share of revenue from 2.5D/3D integration, fan-out WLP, CoWoS-adjacent services vs. legacy wire-bond/flip-chip.

Sources: Segment disclosures (10-K/10-Q), earnings commentary, investor presentations.

Key Evidence:

- Explicit revenue breakout for advanced packaging segments
- Management discussion of packaging mix shift
- Revenue tied to HBM packaging, interposers, fan-out

2.2 Signal B: Capacity-Constrained Positioning

Definition: Sold-out utilization, customer allocation language, lead-time extensions, pricing power evidence.

Sources: Management guidance, order backlog trajectory, 8-K announcements.

Key Language:

- “Sold out”, “fully utilized”, “allocation to key customers”
- “Lead time extension”, “backlog visibility”
- “Price increases”, “premium pricing”, “mix improvement”

2.3 Signal C: AI-Substrate Exposure

Definition: ABF or SAP substrate revenue tied to AI GPU/ASIC/HBM packaging, layer-count complexity, panel size increases.

Sources: Substrate shipment mix, customer concentration, patent activity.

Indicators:

- High layer-count ABF/SAP programs (>12 layers)
- Large panel size production (510mm×515mm+)
- Glass/TGV substrate qualification programs

2.4 Signal D: Test Intensity Leverage

Definition: Growing test time per AI device from larger die, HBM stack validation, and known-good-die (KGD) screening requirements.

Sources: Test revenue per wafer, equipment attach rates, SEMI equipment billings.

Indicators:

- Test segment revenue growth rate

- Mentions of increased test time, KGD screening
- Wafer-level and final test capacity expansion

2.5 Signal E: Next-Gen Packaging Optionality

Definition: Glass core/TGV qualification programs, CPO module integration, hybrid bonding, or embedded passive R&D positioning for 2027–2028 volume ramp.

Sources: Patent filings (CPC-filtered), customer qualification disclosures, capex allocation.

Indicators:

- Patent families in glass/TGV, hybrid bonding, CPO
- Announced pilot lines or sample shipments
- Capex dedicated to next-gen packaging lines

2.6 Signal F: Margin Expansion from Mix Shift

Definition: Gross margin uplift as advanced-packaging revenue displaces commoditized legacy services.

Sources: Margin trend analysis, revenue mix evolution.

Indicators:

- YoY gross margin improvement
- Management commentary on mix-driven margin expansion
- Co-occurrence of “gross margin” with “mix” or “advanced packaging”

2.7 Signal G: Avoid Criteria

Definition: Exclusion flags for immaterial or undifferentiated exposure.

Triggers:

- AI packaging revenue <10% of sales with no credible ramp
- Pure-play front-end foundry (outsources packaging entirely)
- Peak-cycle multiples (EV/Revenue >12×) without capacity guidance
- Commodity materials without differentiation

3 Exposure Scoring Dimensions

3.1 Dimension 1: Advanced Packaging Mix ($w = 0.35$)

Definition: Percent of company revenue derived from advanced packaging services or products (2.5D/3D, CoWoS, fan-out WLP, ABF/SAP substrate sales, CPO modules) vs. legacy wire-bond/flip-chip.

Measurement Fields:

- `segment_revenue_pct_packaging` (from 10-K/10-Q)
- `company_overview.topic_relevance_answer`
- `earnings_transcript_mentions_count` of advanced packaging terms
- explicit IR revenue breakout

Missing Data Rule: If explicit revenue breakout is missing, estimate using text classification to assign packaging-likelihood score and scale by total revenue; mark higher uncertainty flag. If estimated share <10%, apply automatic penalty.

3.2 Dimension 2: Capacity Constraint Signal ($w = 0.30$)

Definition: Quantified evidence that the firm’s advanced-packaging capacity is sold out / constrained and commanding price or allocation power.

Measurement Fields:

- Counts/intensity of “sold out”, “lead time”, “backlog”, “allocation” mentions
- Narrative trend (increasing/decreasing) of backlog across last 4 quarters
- Order backlog announcements or management guidance flags

Missing Data Rule: If numeric backlog/lead-time figures absent, use normalized count-based signal (mentions per 10K words) with LLM-derived intensity weighting.

3.3 Dimension 3: AI Substrate Complexity ($w = 0.20$)

Definition: Exposure to complex AI-focused substrate measured by revenue linkage, patent activity, and presence of high-layer-count or large-panel substrate programs.

Measurement Fields:

- patent_count_filtered_by_relevant_CPC/IPC (5-year window)
- patent_qualitative_relevance_score
- company statements of ABF/SAP/glass substrate qualification
- segment_revenue_pct_from_substrates

Missing Data Rule: If company-level substrate revenue not disclosed, rely on patents per employee or patent relevance ratio and text mentions.

3.4 Dimension 4: Test Intensity Leverage ($w = 0.15$)

Definition: Degree to which the company benefits from rising test intensity per AI device.

Measurement Fields:

- test_segment_revenue_growth_rate
- mentions of increased test time, KGD screening, wafer-level test
- SEMI equipment billings trend (contextual normalization)

Missing Data Rule: If firm-level test attach metrics unavailable, use combined signal of test-segment revenue trend and text-mention intensity.

4 Composite Score Construction

4.1 Exposure Scoring Composite Formula

$$\text{Score}_i = 0.35 \cdot z(\text{APM}_i) + 0.30 \cdot z(\text{CAP}_i) + 0.20 \cdot z(\text{SUB}_i) + 0.15 \cdot z(\text{TEST}_i) - 1.0 \cdot \mathbf{1}(\text{APM_pct}_i < 10\%) \quad (1)$$

where APM = Advanced Packaging Mix, CAP = Capacity Constraint Signal, SUB = AI Substrate Complexity, TEST = Test Intensity Leverage.

4.2 Factor Methodology Composite Formula

$$\text{Score} = 0.30 \cdot \text{APM} + 0.25 \cdot \text{CAP} + 0.15 \cdot \text{SUB} + 0.10 \cdot \text{TEST} + 0.10 \cdot \text{NEXT} + 0.10 \cdot \text{MARGIN} \quad (2)$$

4.3 Scoring Dimension Formulas

Dimension	Formula	Weight
Advanced_Packaging_Mix	$0.5 \cdot z[K(\text{pkg_terms})] + 0.3 \cdot R_{\text{topic}} + 0.2 \cdot (-z[K(\text{legacy})])$	0.30
Capacity_Constraint	$0.6 \cdot z[K(\text{cap_terms})] + 0.4 \cdot z[\Delta K(\text{cap_terms})]$	0.25
AI_Substrate_Complexity	$0.5 \cdot \text{pctl}(\log(1 + \text{patents}_{5y})) + 0.5 \cdot z[K(\text{sub_terms})]$	0.15
Test_Intensity	$0.7 \cdot z[K(\text{test_terms})] + 0.3 \cdot R_{\text{test_topic}}$	0.10
NextGen_Optionality	$0.6 \cdot \text{pctl}(\log(1 + \text{patents}_{5y})) + 0.4 \cdot z[K(\text{qual_terms})]$	0.10
Margin_Mix_Expansion	$\text{cooccurrence}(\text{"gross margin"}, \text{"mix/adv pkg"}) \cap$	0.10

Table 2: Scoring dimension formulas

where $K(\cdot)$ = keyword density per 10K words, R_{topic} = topic relevance score, pctl = percentile transform.

4.4 Normalization Method

1. Compute z-scores per dimension across the final filtered universe
2. Winsorize: Clip each z-score to $[-3, +3]$
3. Percentile Transform: Map clipped z to percentile $[0, 1]$ via empirical CDF
4. Weighted Sum: Apply dimension weights to percentiles
5. Missing Values: Impute median percentile (0.5) and apply uncertainty penalty

4.5 Score Interpretation

Percentile	Interpretation
p90–p100	Core long — high conviction exposure to constrained packaging tailwinds
p70–p90	Satellite long — clear exposure but with execution/visibility caveats
p40–p70	Watchlist / neutral — partial or uncertain exposure; monitor disclosures
Below p40	Exclude / avoid — minimal differentiated AI-packaging exposure (<10%)

Table 3: Score interpretation by percentile

5 Portfolio Weight Construction

5.1 Base Weight Formula

$$w_i = \frac{\max(\text{Score}_i, 0)^{1.5}}{\sum_j \max(\text{Score}_j, 0)^{1.5}} \quad (3)$$

The exponent $p = 1.5$ provides modest convexity to overweight higher-scoring names.

5.2 Risk Controls and Caps

Constraint	Value
Maximum Single-Name Weight	3%
Minimum Position Weight	0.05%
Turnover Budget (monthly)	20% notional
Drawdown Guardrail	Halve top bucket exposure after -12% drawdown
Stop Loss (single-name)	20% trailing from entry
Annualized Turnover Target	120%

Table 4: Portfolio risk controls

5.3 Role Bucket Caps

Role Bucket	Maximum Weight
OSAT / Advanced Packaging	30%
Substrates (ABF/SAP/Glass)	25%
Test Equipment & Services	20%
Packaging Equipment & Materials	15%
Embedded Passives & Optics	10%

Table 5: Role bucket weight caps

5.4 Application Order

1. Drop names with $\text{Score}_i \leq 0$ or failing tradability filters
2. Compute base weights via power-law formula
3. Apply role bucket caps; redistribute excess pro-rata within other buckets
4. Apply 3% single-name cap; redistribute excess
5. Renormalize to sum to 100%

6 Tradability Filters

Filter	Threshold	Source
Minimum 30-day ADV	$\geq \$5,000,000$	<code>polygon_adapter.avg_daily_volume</code>
Minimum Price	$\geq \$5.00$	<code>polygon_adapter.get_data</code>
Minimum Market Cap	$\geq \$250,000,000$	Market data
Minimum Free Float	$\geq 10\%$	Market data
Listing Venues	NYSE, NASDAQ, AMEX	Exchange data
Excluded Types	OTC, Pink Sheets, ETFs, Preferreds	Listing classification
Corporate Actions	Exclude pending delisting, merger cash-out, bankruptcy	Filings

Table 6: Tradability filter thresholds

6.1 Valuation Sanity Filter

Exclude names with $\text{EV} / \text{trailing-12m revenue} > 12\times$ unless:

- Advanced-packaging revenue share $> 25\%$, AND
- Capacity guidance indicates multi-quarter under-supply

7 Classification Tiers

Tier	Score Range	Weight Range	Characteristics
Tier 1: Pure-Play Enablers	[1.0, 3.0]	[2%, 3%]	Direct packaging/substrate/test revenue
Tier 2: Significant Enablers	[0.3, 1.0]	[1%, 2%]	Material exposure, capacity evidence
Tier 3: Tangential/Optional	[0.0, 0.3]	[0.5%, 1%]	Limited direct exposure, optionality

Table 7: Classification tiers by composite score

8 Data Requirements

8.1 Adapter Data Requirements

8.1.1 `sec_adapter` (MUST HAVE)

- `get_filings`: 10-K, 10-Q, 8-K text content
- Fields: `segment_revenue_advanced_packaging_pct`, `margin_by_segment`, `capacity_and_capex`, `backlog_and_utilization_text`
- Freshness: 30 days

8.1.2 `fmp_adapter` (MUST HAVE)

- `get_earnings_transcripts`: Earnings call transcripts
- Fields: `backlog_and_utilization`, `pricing_power_phrases`, `customer_allocation_language`
- Freshness: 14 days

8.1.3 `patents_adapter` (MUST HAVE)

- `get_patents_by_tickers_batch`: Patent records by CPC/IPC
- Fields: `patent_counts_by_technology` (`glass_core`, `TGV`, `hybrid_bonding`, `CPO`, `embedded_passives`)
- CPC codes: H01L (semiconductors), C03 (glass), B23K (bonding), H05K (substrates), G01R (test)
- Lookback: 5 years

8.1.4 `company_overview` (MUST HAVE)

- `get_company_profile`: Industry, sector, description
- `check_topic_relevance`: Packaging/substrate/test relevance

8.1.5 `polygon_adapter` (MUST HAVE)

- `get_daily_close_prices`: Price history
- `avg_dollar_volume`: 30-day ADV
- `get_short_interest`: Days-to-cover

8.2 External Sources

8.3 Missing Data Requirements

1. Detailed substrate shipment mix (ABF vs. SAP vs. glass) by customer — often only in vendor IR decks
2. ASPs by substrate/material SKU at high frequency — paywalled industry reports

Domain	Data Types
Company IR sites	IR decks, press releases, capex announcements
semi.org	Equipment billings, industry outlooks
imarcgroup.com	/ Materials pricing (epoxy, underfill, mold compound)
dataintel.com	
patents.google.com	/ Patent full text, claims, classification
epo.org	
Standards bodies (JEITA, IPC, JEDEC)	Qualification standards, test specs

Table 8: External data sources

3. Factory-level utilization percentages by substrate type — partially via exhibits/IR statements
4. Customer-level attach rates for test (test_time_per_device) — must be proxied

9 Keyword Dictionaries

9.1 Packaging Terms ($K_{\text{packaging}}$)

- advanced packaging, CoWoS, CoPoS, 2.5D, 3D integration
- fan-out, fan-out WLP, FOWLP, HBM package
- interposer, silicon interposer, organic interposer
- hybrid bonding, copper pillar, micro-bump
- substrate, ABF, SAP, glass core, TGV
- underfill, mold compound, bonding film, flux

9.2 Legacy Terms (K_{legacy})

- wire bond, wire bonding, flip-chip (without advanced qualifier)
- legacy packaging, traditional packaging
- commoditized assembly

9.3 Capacity Terms (K_{capacity})

- sold out, fully utilized, capacity constrained
- allocation, customer allocation, prioritized customers
- lead time, lead-time extension, backlog
- price increase, pricing power, premium pricing
- order book, booking, book-to-bill

9.4 Substrate Terms ($K_{\text{substrate}}$)

- ABF substrate, SAP substrate, build-up substrate
- glass substrate, glass core, TGV, through-glass via
- high layer count, large panel, panel-level
- AI substrate, GPU substrate, HBM substrate

9.5 Test Terms (K_{test})

- wafer test, final test, package test

- test time, test intensity, known-good-die, KGD
- ATE, automatic test equipment
- burn-in, reliability test, HBM test

9.6 Qualification / Next-Gen Terms ($K_{\text{qualification}}$)

- qualification, customer qualification, qual program
- pilot line, sample shipment, production ramp
- glass qualification, TGV qualification
- CPO, co-packaged optics, silicon photonics
- hybrid bonding, direct bonding

10 Execution Pipeline

10.1 Phase 1: Universe Construction (Steps 1–5)

Step	Name	Agent	Description
1	load_base_universe	UniverseSelectionAgent	S&P 500 + Russell 2000
2	apply_sector_filters	UniverseSelectionAgent	Filter to target industries
3	pull_company_profiles	GenericExtractionAgent	company_overview.get_profile
4	topic_relevance	GenericExtractionAgent	check_topic_relevance
5	role_tagging	ClassificationAgent	Assign role bucket labels

Table 9: Phase 1 execution steps

10.2 Phase 2: Data Collection (Steps 6–9)

Step	Name	Agent	Description
6	fetch_sec_filings	GenericExtractionAgent	10-K/10-Q/8-K for 3 years
7	fetch_transcripts	GenericExtractionAgent	Earnings transcripts (8 quarters)
8	fetch_patents	GenericExtractionAgent	Patents by CPC (5-year window)
9	fetch_prices_liquidity	GenericExtractionAgent	OHLCV, ADV, short interest

Table 10: Phase 2 execution steps

10.3 Phase 3: Feature Extraction (Steps 10–12)

Step	Name	Agent	Description
10	classify_text	ClassificationAgent	Label paragraphs for signals A–F
11	derive_structured_features	GenericExtractionAgent	Extract numeric metrics
12	compute_patent_signals	GenericExtractionAgent	Patent counts by topic

Table 11: Phase 3 execution steps

Step	Name	Agent	Description
13	score_dimensions	GenericScoringAgent	z-scores, percentiles, composite
14	apply_eligibility_filters	CodeGeneratingAgent	$\geq 10\%$ APM, tradability
15	apply_tradability	CodeGeneratingAgent	ADV, price, exchange filters
16	position_sizing	CodeGeneratingAgent	Power-law weights + caps
17	output_rebalance_pack	OutputAgent	Weights CSV, audit trail, HTML

Table 12: Phase 4 execution steps

10.4 Phase 4: Scoring & Portfolio Construction (Steps 13–17)

11 Rebalancing Rules

11.1 Schedule

- **Frequency:** Monthly on the first US business day
- **Announcement Lag:** Publish T–2 business days pre-effective date
- **Data Cutoff:** Frozen membership unless corporate actions necessitate changes

11.2 Buffer Rules

- **Rank Buffer:** Add only if composite rank enters top 70th percentile; drop only if rank falls below 60th percentile
- **Role Buffer:** If role changes between adjacent categories, require confirmation for one additional month unless Score changes by >0.5 SD

11.3 Turnover Control

- **Budget:** 20% notional per month; defer excess to subsequent cycle
- **Cash Handling:** Residual cash from capping/rounding held in cash proxy or pro-rata across holdings

12 Risk and Validation

12.1 Diagnostics

Diagnostic	Description
lead_time_trend	Time-series of lead-time statements; trend slope and last-quarter delta
backlog_months_trend	Order backlog normalized to quarterly revenue (3-, 6-, 12-month changes)
ASP_trend	Price per-unit proxies for ABF/SAP, underfill, mold compound (QoQ, YoY)
hiring_intensity	Engineering/manufacturing headcount and capex hiring normalized YoY

Table 13: Diagnostic metrics

12.2 Validation Tests

1. **Window:** Rolling 12-month lookback; 1-, 3-, 5-year validation windows

2. **Benchmark:** Compare vs. IT – Semiconductors sector and broad-semi ETF
3. **Top Bucket Hit Rate:** Fraction of top 20 names that subsequently report >10% YoY packaging revenue
4. **IC Mean:** Rank IC vs. 3-, 6-, 12-month forward margin and revenue surprise; report mean and t-stat
5. **Capacity News Hit Rate:** Rate of confirming capacity/backlog news within 90 days of selection
6. **Drawdown Stats:** Maximum drawdown and recovery time; stress test under capacity release scenarios

12.3 Validation Criteria

- Coverage: $\geq 85\%$ of eligible universe has non-imputed scores on at least 4 of 6 dimensions
- Cross-sectional IC: Average monthly IC $\geq 2\%$ with $p < 0.10$ over 3-year backtest
- Hit rate: Top-quintile Score outperforms bottom-quintile by ≥ 300 bps annualized
- Liquidity resilience: Simulated trading executes $\leq 10\%$ of constituent-level ADV per day
- Capping compliance: No breaches of 3% name cap and role bucket caps

12.4 Failure Modes

1. Double-counting front-end foundries that merely resell packaging services
2. Immaterial AI packaging revenue (<10%) incorrectly treated as core exposure
3. Cyclical peak with no durable capacity lead-times (short-lived pricing)
4. Commodity materials providers without differentiation treated as strategic
5. Export-control shocks disrupting supply/demand for advanced-packaging tools

13 Sample Companies

Ticker	Name	Role / Relevance
AMKR	Amkor Technology	OSAT: advanced packaging, fan-out, 2.5D/3D
ASX	ASE Technology	OSAT: CoWoS capacity, HBM packaging
TER	Teradyne	Test equipment: ATE for advanced packaging
KLAC	KLA Corporation	Inspection/metrology: packaging inspection
KLIC	Kulicke & Soffa	Packaging equipment: bonding, advanced assembly
COHR	Coherent Corp	Substrates, materials: SiC, optical components
–	Ibiden	Substrates: ABF substrates for AI/HPC

Table 14: Sample companies (illustrative; actual inclusion via scoring pipeline)

14 Implementation Notes

14.1 Text Parsing

- Lowercase, remove punctuation/numbers, stop-word removal

- Compute keyword densities per 10K words for stability across filing lengths
- Co-occurrence uses a 50-word symmetric window

14.2 Patent Mapping

- Use GenericExtractionAgent to infer CPC/IPC families relevant to:
 - ABF/SAP: H05K (printed circuits, substrates)
 - Glass/TGV: C03C/G02B (glass, optical)
 - CPO/Optical: G02B/H01S (optical integration)
 - Embedded Passives: H01L (semiconductor devices)
- Review top-N CPCs by frequency per role and lock definitions for backtest window

14.3 Weighting Robustness

- Fixed, interpretable weights avoid unstable optimized blends
- Aligns with practitioner guidance on multi-signal weighting
- Avoids overfitting per “factor zoo” research cautions

14.4 Auditability

- Retain per-ticker JSON with term counts, snippets (± 100 chars around matched terms)
- Store CPC lists and score components for review
- Link each score to source filings/patents for traceability

15 Citations

15.1 Industry Sources

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2. FTSE Russell Capping Methodology Guide (v5.0, Jan 2026). https://www.lseg.com/content/dam/ftse-russell/en_us/documents/policy-documents/capping-methodology-guide.pdf
3. FTSE GEIS – Guide to Calculation Method for Median Liquidity Test (Feb 2026). https://www.lseg.com/content/dam/ftse-russell/en_us/documents/policy-documents/guide-to-calculation.pdf
4. MSCI Index Calculation Methodology (Feb 2026). https://app2.msci.com/eqb/methodology/meth_docs/MSCI_Index_Calculation_Methodology_Feb2026.pdf

15.4 Academic References

1. Kozak/Nagel/Santosh (2017): Shrinkage Across Many Predictors. NBER WP 24070. <https://www.nber.org/papers/w24070>
2. Feng/Giglio/Xiu (2020): Taming the Factor Zoo. https://dachxiu.chicagobooth.edu/download/Z00_SSRN.pdf
3. FactSet Insight: A Practical Approach to Weighting Signals (Apr 2026). <https://insight.factset.com/a-practical-approach-to-weighting-signals>
4. Kelly et al. (2018): Patent Text as Innovation Indicator. NBER WP 25266. https://www.nber.org/system/files/working_papers/w25266/revisions/w25266.rev0.pdf

A Appendix A: Extraction Metadata

Field	Value
Extraction Timestamp	2026-04-29T15:52:29.998652+00:00
Pipeline Version	4.0.0
Documents Processed	227
Total Snippets Extracted	473
High-Confidence Threshold	0.5
Snippets After Filtering	120
Classification Model	gpt-5-mini
Agent Extraction Model	gpt-5-mini
Overall Confidence (Universe/Data)	0.82
Overall Confidence (Exposure Scoring)	0.78
Overall Confidence (Factor Methodology)	0.78
Overall Confidence (Risk/Validation)	0.78

Table 15: Extraction metadata

B Appendix B: CPC Code Mappings

C Appendix C: Step-by-Step Implementation Guide

1. **Load Base Universes:** Use UniverseSelectionAgent to load S&P 500 and Russell 2000 constituents.
2. **Apply Industry Filters:** Keep Information Technology → Semiconductors & Equipment; IT → Electronic Equipment; Materials → Specialty Chemicals.
3. **Pull Company Profiles:** Call company_overview.get_company_profile and check_topic_relevance with topic = “advanced packaging OR OSAT OR substrate OR ABF OR SAP OR glass core OR TGV OR CPO OR wafer test OR package test OR underfill OR mold compound”.

Technology	Relevant CPC Codes
Semiconductor Devices	H01L (general)
Glass Processing	C03 (glass compositions, surface treatment)
Bonding/Joining	B23K (laser, ultrasonic, thermocompression)
Substrates/PCBs	H05K (printed circuits, mounting)
Test/Measurement	G01R (measuring, testing apparatus)
Optical Components	G02B (optical elements, systems)
Laser/Photonics	H01S (devices using stimulated emission)

Table 16: CPC code mappings for patent filtering

4. **Role Tagging:** ClassificationAgent labels each ticker into role buckets using profile descriptions.
5. **Acquire SEC Texts:** sec_adapter.get_filings for form_types = [10-K, 10-Q, 8-K] over last 365 days.
6. **Compute Packaging Mix Features:** CodeGeneratingAgent parses text_content; compute keyword densities per 10K words for packaging and legacy terms.
7. **Compute Capacity Features:** Extract capacity/constraint language; compute mention counts and QoQ trends.
8. **Compute Substrate Features:** Combine patent counts (5-year, CPC-filtered) with substrate keyword density.
9. **Compute Test Features:** Extract test-intensity language and equipment billings proxies.
10. **Compute Next-Gen Features:** Patent counts for glass/TGV/CPO/hybrid bonding + qualification term density.
11. **Composite Score:** GenericAggregationAgent combines dimensions using fixed weights; apply penalties.
12. **Tradability Filters:** Enforce $ADV \geq \$5M$, $price \geq \$5$, exchange whitelist; keep only eligible names.
13. **Position Sizing:** CodeGeneratingAgent calculates $w_i = \max(\text{Score}_i, 0)^{1.5} / \sum$; apply role bucket caps and 3% max-name cap.
14. **Rebalance:** Monthly on first business day; apply 20% turnover budget and rank buffers; publish T-2 pre-trade files.
15. **Monitoring:** Track lead-time/backlog trends, drawdown guardrails, attribution by role bucket; store audit trail.